THE BITS HYDERABAD CONSULTING GROUP

BHCG

BREAKTHECASE'26

CASE STUDY COMPETITION

Green Energy GCC Consulting

Reducing Invoice Processing Defects through a Data-Driven Six Sigma Approach



TEAM

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1 DEFINE | 2 MEASURE | 3 ANALYZE | 4 IMPROVE | 5 CONTROL

Problem & Context



Why it got worse
(The complication)

Portfolio expansion
The renewable portfolio expanded recently, so procurement and invoice volume jumped.

Multiple vendors and formats
Invoices come from many regions and vendors, each with its own format, payment terms and approval flow.

Volume fluctuations
Volume swings a lot. A normal day is about 300 invoices, but month-end peaks go as high as 880.

The Problem

A green energy company's GCC processes about **350 invoices a day** across solar, wind and infrastructure projects.

Each invoice form has **58 fields**, and wrong or missing entries keep recurring.

Result: delayed payments, unhappy vendors, and **GST penalties** of up to **₹5,000 per invoice.**

Business Impact
(Baseline Numbers)

1,669
Invoices audited in baseline
(10% of 16,663 in 52 working days)

990
Wrong fields found
out of 96,802 checked

1.02%
Error rate of fields
(0.59 errors per invoice)

₹10.4 lakh
Estimated cost of errors per month

Timeline (High-level Plan)

1. **Baseline (60 days)**
Collect data on recurring defects across the 58 fields.

2. **Root cause analysis (within 2 months)**
Identify and validate root causes with a 10% sample audit every day.

3. **Implementation (6 months)**
Put the improvement actions in place across the process.

Data Validation

1 Validation checks

Totals tie
Measure errors (990) equal the Error Reasons total (990).

Calendar
52 working days, 1 Aug to 30 Sep 2026, no duplicate dates, no Sundays (matches the Glossary).

Sample size
Audited invoices equal 10% of received, rounded, on all 52 days; overall 10.02% (1,669 of 16,663).

Errors above samples
31 Aug (48 of 46), 3 Sep (28 of 27), 28 Sep (92 of 87): valid; one invoice can carry several errors.

2 The 25 Aug anomaly

25 AUGUST

659 invoices received, 66 audited, 0 errors.

The only zero-error day among 52 working days; it falls at the start of the month-end peak, between a low-error day (24 Aug, 0.19) and a high-error day (26 Aug, 0.80).

Errors per audited invoice



3 Volume and period checks

Baseline (1 Aug – 30 Sep 2026)

52	1,669	16,663
working days	audited invoices (10.02%)	received invoices

Daily / monthly volume
320 invoices a day, 8,332 a month; inside the case range of 7,000 to 9,100 (Aug 7,604, Sep 9,059).

Improve period (1 Mar – 28 Apr 2027)

51 working days

Same rounding rule
No anomalies (maximum 6 errors a day).

4 Treatment sensitivity — all remain ~3.8 sigma

Treatment	Audited invoices	Errors	DPMO	Sigma	Notes
As recorded	1,669	990	10,227	3.82	Includes 25 Aug (0 errors).
Excluded (used in this deck)	**1,603**	**990**	**10,648**	**3.80**	Drop 25 Aug (66 audited invoices) from error-rate denominator.
Imputed at peak-day rate	1,669	1,028	10,618	3.80	Assume 0.57 errors per audited invoice on 25 Aug (about 38 errors).

Source: Measure, Improve, Glossary and Error Reasons tabs. Imputation uses 0.573 errors per audited invoice × 66 ≈ 38 errors.

Baseline and sigma



1 Key baseline numbers

16,663
Invoices received (320 a day)

1,603
Invoices audited (validated)

990
Errors found

2 What do these numbers mean?

62
errors per 100 invoices

1
wrong box in every 100 field checks (each form has 58 boxes)

3 From errors to DPMO and sigma

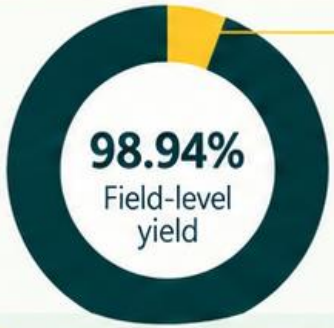
0.62
Errors per audited invoice

→ **1.06%**
Defect rate per opportunity

→ **10,648**
DPMO

→ **3.80**
Sigma level

4 Field-level yield



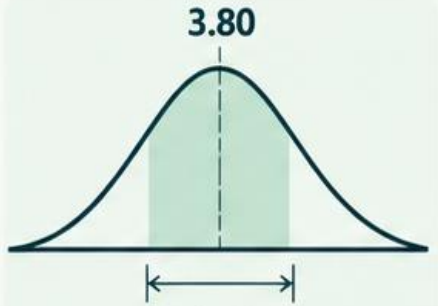
1.06%
Defect rate (errors)

Out of every 100 field checks, about 99 are correct and 1 is wrong. Each form has 58 boxes, so about 1 box in 100 is wrong.

5 Sigma ladder

Sigma level	DPMO
3.0	66,807
3.80 (today)	**10,648**
4.0	6,210
5.0	233
6.0	3.4

6 Confidence in the baseline



Sampling range (95%)

3.78 to 3.83 sigma

(based on Poisson range on 990 errors)

Application of Six Sigma (DMAIC)



5 DMAIC flow for this project

D Define

What is wrong, for whom, within what limits?

0.62
errors per invoice (58 fields)

Gate: scope agreed with CFO & Procurement Head

M Measure

How bad is it, and can we trust the data?

10,648 DPMO
3.80 sigma

Gate: validated baseline

A Analyze

Why does it happen and what does it cost?

Top 3 = 75.3%
of 990 errors
Vendor origin = 94.9%

Gate: top 3 root causes confirmed on data

I Improve

What removes the causes, and how do we prove it?

435 DPMO
4.83 sigma

Gate: at least 90% DPMO cut

C Control

How do we keep it?

0.025
errors per invoice (30 audited invoices)

Gate: sigma held at 4.5 or above

The big picture



Timeline with phases and goals

Define	Measure	Analyze	Improve	Control
Aug – Sep 2026	Aug – Sep 2026	Aug – Sep 2026	Oct 2026 – Feb 2027	Mar – Apr 2027 onwards
Set scope and goal (90% DPMO cut)	Validate data and find baseline	Find root causes and quantify cost	Implement fixes and prove results	Monitor daily and sustain gains

How DMAIC handles our four complications

Complication	Where DMAIC deals with it	What we do
GST penalties of up to ₹5,000 per invoice	Define, Analyze	State it in the problem; size it at ₹7.29 lakh a month of the ₹9.33 lakh
94.9% of errors start with vendors	Analyze, Improve	Fix at the point of submission instead of checking harder downstream
Month-end peaks	Measure, Control	Show the pattern; add a month-end trigger to the control plan
Data and sampling risk (25 Aug, 10% audit)	Measure	Validate, state the treatment, report sigma with its range (3.78 to 3.83 before, 4.74 to 4.92 after)

Our approach in five rules

1. **Data first:** nothing is claimed that the workbook cannot show.
2. **Fix at the source:** the vendor, not the analyst, is where errors start.
3. **Pilot before scale:** prove each fix on a few vendors before rolling out.
4. **Prove with a second audit:** same 58 fields, same 10% sampling, same formula.
5. **Control with a chart:** the daily audit becomes the early-warning system.

Month-end Patterns



1 Peak vs other days

Share of invoices

Peak days (late month)	62.7%
Other days (1–24)	37.3%

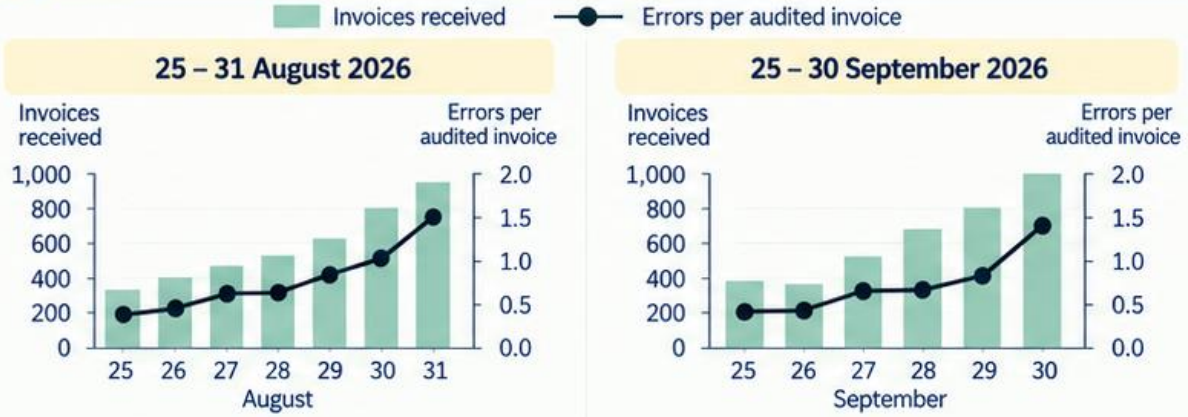
Share of errors

Peak days (late month)	69.3%
Other days (1–24)	30.7%

2 Daily trend: invoice volume and error rate



3 Month-end comparison

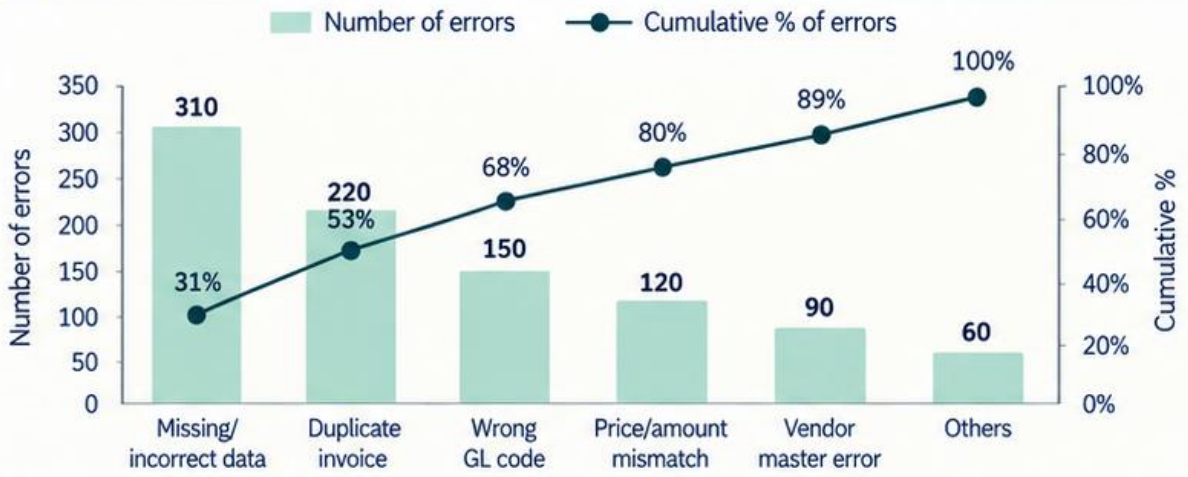


4 Key takeaways

1. Month-end days process **62.7%** of all invoices but account for **69.3% of total errors.**
2. Error rate on peak days **(0.70) is higher** than on other days **(0.49).**
3. Both August and September show a clear spike in errors in the last week, with **September's peak higher than August's.**
4. High workload at month-end is **associated with higher error rates,** indicating a need for targeted interventions.

Pareto Analysis: A few error types drive most defects

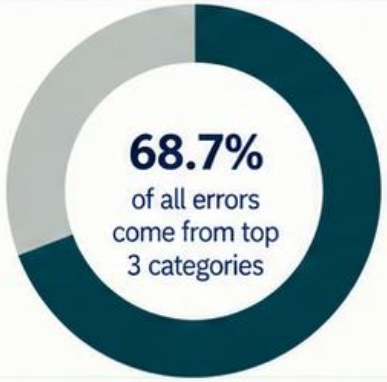
1 Pareto chart – error categories



2 Top error categories by impact

Rank	Error category	Count of errors	% of total errors	Cumulative %
1	Missing/incorrect data	310	31.3%	31.3%
2	Duplicate invoice	220	22.2%	53.5%
3	Wrong GL code	150	15.2%	68.7%
4	Price/amount mismatch	120	12.1%	80.8%
5	Vendor master error	90	9.1%	89.9%
6	Others (10+ categories)	60	6.1%	100.0%

3 Focus on the vital few



Top 3 categories
(310 + 220 + 150 = 680 errors)
68.7%

Remaining 3 categories
(120 + 90 + 60 = 270 errors)
31.3%

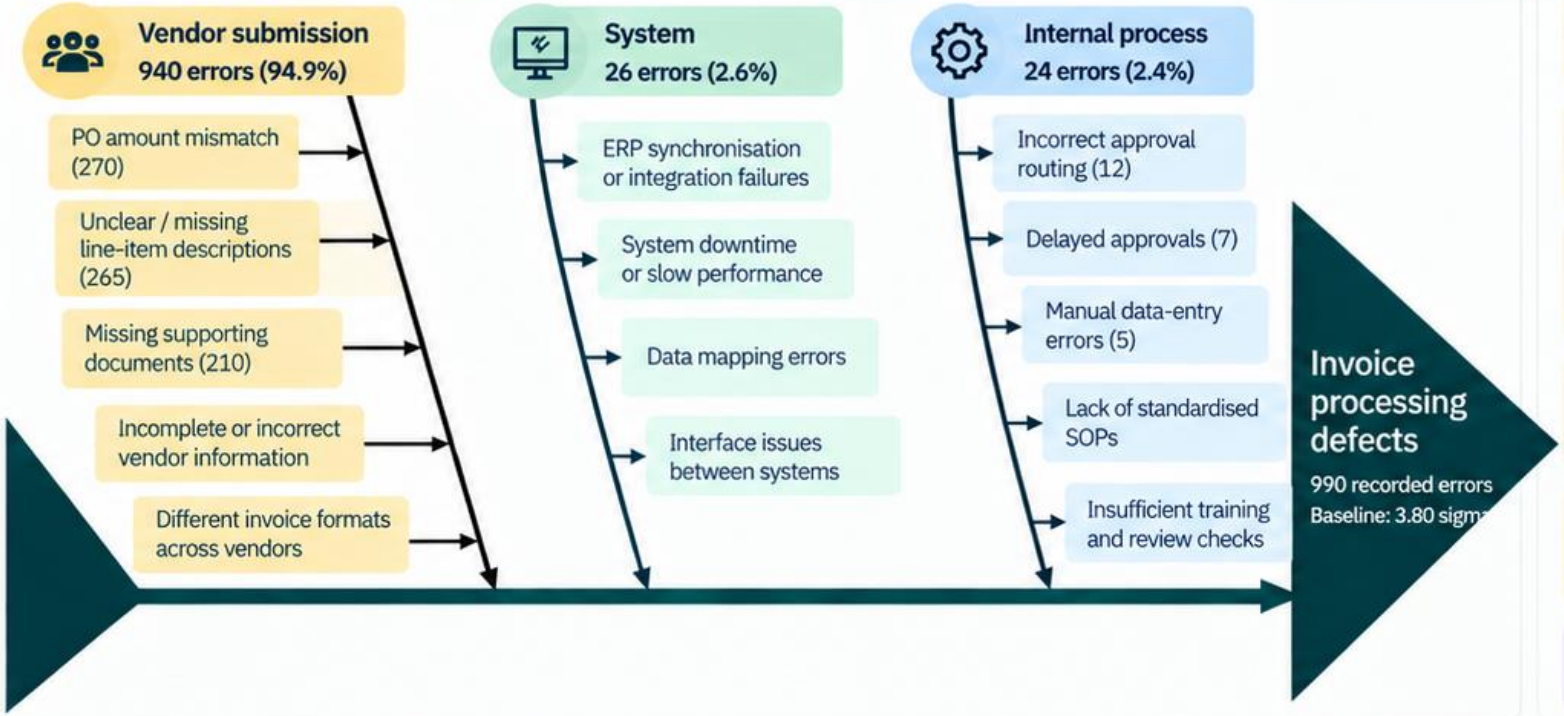
4 Key takeaways

1. The top 3 error categories (missing/incorrect data, duplicate invoice, wrong GL code) account for **68.7%** of all **990 errors**.
2. Missing/incorrect data alone contributes **31.3%** of errors, making it the largest single driver.
3. Focusing on the top 3 categories can deliver the **biggest reduction** in errors with targeted interventions.
4. The remaining categories contribute **31.3%** and can be addressed in later phases.

Root Causes of Invoice Defects



1 Root cause analysis



2 5 Whys – Example: Invoice and PO amount mismatch

1	**What happened?**	Invoice amount did not match the purchase order. **(270 errors)**
2	**Why might this have happened?**	Vendors may prepare invoices separately from PO information.
3	**Why might this be happening?**	Price, quantity or change-order differences may not be visible during invoice preparation.
4	**Why does this continue to happen?**	There may be no automated PO-linked validation before submission.
5	**Root-cause hypothesis to validate**	Is the absence of an effective matching check allowing mismatches into the process?

Note: These are potential causes based on data and process understanding, to be validated through further analysis and stakeholder inputs.

3 Key takeaway

Prevention should begin at submission: validate **PO amounts, standardise invoice descriptions and require supporting documents.**

Process Bottlenecks

1 Invoice processing flow and where defects occur

1. Receipt & logging	2. Format and field validation	3. Vendor verification	4. PO matching	5. Approval routing	6. Exception handling	7. GST / tax validation	8. Final approval & payment
Capture invoice in system	Check required fields, format and data accuracy	Validate vendor details and master data	Match invoice with purchase order (PO)	Route to approvers based on value and category	Resolve mismatches and missing information	Validate GST details and tax compliance	Final review and release for payment
	265 errors (26.8%)	90 errors (9.1%)	270 errors (27.3%)	24 errors (2.4%)	26 errors (2.6%)	12 errors (1.2%)	Others (28 errors, 2.8%)

2 Defect concentration across the process



635 of 990 errors (64.1%) occur in the early stages of the process, mainly due to PO amount mismatches and incomplete or unclear invoice descriptions.

3 Capacity pressure during peak periods

375
Invoices/day
Regular processing capacity

653
Invoices/day
Average peak-day volume

880
Invoices
Busiest day observed

17 of 52 days
Above regular capacity

9 of 52 days
Above overtime capacity

Key insight

Peak-day demand (average 653 invoices) exceeds the regular capacity (375) by ~74%, and even surpasses overtime capacity on 9 days. This increases the risk of errors, especially in early validation and PO matching.

The Cost of Invoice



1 Total estimated cost of invoice errors

₹ 9.33 lakh per month

₹ 1.12 crore per year

Based on 990 errors in the baseline period and current processing volumes. Includes GST penalty exposure and analyst rework effort.

2 Breakdown of monthly and annual cost

Cost component	Basis for estimation	Monthly cost (₹ lakh)	Annual cost (₹ lakh)
GST penalty exposure	• 270 invoices with amount mismatches • Penalty rate: assumed 2% of invoice value • Average mismatch value: ₹1.35 lakh	7.29	87.5
Analyst rework cost	• 990 errors × 15 minutes of rework each • Analyst cost: ₹500 per hour • Rework cost per error: ₹125	2.04	24.5
Total estimated cost		**9.33**	**112.0**

3 Monthly cost breakdown



GST penalty exposure
₹ 7.29 lakh

Analyst rework cost
₹ 2.04 lakh

4 Cost per error (average)

Average penalty cost per error

₹ 2,700

(2% of ₹1.35 lakh mismatch value)

Average rework cost per error

₹ 125

(15 minutes × ₹500/hour)

5 Sensitivity: What if assumptions change?

Average penalty per mismatched invoice	Monthly cost (₹ lakh)	Annual cost (₹ lakh)
Base case (2%)	**7.29**	**87.5**
Lower case (1%)	3.65	43.8
Higher case (3%)	10.94	131.3

Even at a lower 1% penalty rate, the annual cost remains ₹68.3 lakh, showing a strong business case for improvement (when including analyst rework cost).

Charter, Risk & Sensitivity Analysis

1 Project Charter

Goal: Cut DPMO by at least 90% (10,648 to 1,065 or below; 3.80 to 4.57 sigma or above) in 6 months.

Business case: Errors cost ₹9.33 lakh/month (₹1.12 crore/year).

Scope: Invoice receipt to payment authorization (steps 1–9); Treasury steps 10–12 excluded.

Roles: Sponsor: CFO. Co-sponsor: Procurement Head. Lead: Six Sigma consultant. Process owner: head of invoice processing.

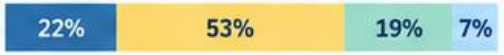
Milestones: Aug–Sep 2026 baseline, Oct 2026–Feb 2027 rollout, Mar–Apr 2027 audit, May 2027 control.

2 Value Chain & Time

Requisition → PO → Delivery → Vendor invoice → Steps 1–9 (our scope) → Payment and archive (Treasury)

Position: a support activity (finance and procurement) serving project delivery.

Where the 91.25 elapsed minutes go



Hands-on work · Waiting for approvals · Other waiting · Rework

Error vs clean invoice processing time

32 vs 20 analyst minutes (+60%)

110 vs 85 elapsed minutes (+29%)

3 Process Risk Analysis

Step	Errors	Risk type	RPN
5 PO matching	304	Vendor (human)	**400**
7 Exception handling	210	Vendor (human)	**378**
1–2 Receipt and logging	75	Vendor 44, machine 26, analyst 5	**336**
3 Field validation	331	Vendor (human)	**280**
4 Vendor verification	36	Vendor (human)	**200**
8 GST and tax	15	Vendor (human)	**189**
6 Approval routing	19	Process	**60**

Risk by type: vendor 94.9%, machine 2.6%, process 1.9%, analyst 0.5%.

Risk reading: Highest risks are where errors are caught late (steps 5, 7, 1–2). Step 3 has the most errors but ranks 4th because it is caught early.

Response: Automated gates at submission (PO match, mandatory fields, attachment check, duplicate check).

4 Sensitivity Analysis

Parameter	Low to high	Monthly saving
Share penalised	2.5% to 7.5%	₹5.45 to 12.45 lakh
Share with an error	25% to 45%	₹6.39 to 11.51 lakh
Penalty per invoice	₹2,500 to ₹5,000	₹5.45 to 8.95 lakh
Invoices per month	7,000 to 9,100	₹7.52 to 9.78 lakh
Reduction achieved	80% to 95.9%	₹7.47 to 8.95 lakh
Exception time	9 to 15 min	₹8.46 to 9.44 lakh
Analyst cost per hour	₹280 to ₹420	₹8.56 to 9.34 lakh

Base monthly saving (at mid-case):
₹8.95 lakh

Lowest single flex: ₹5.45 lakh a month.

All seven at their low values together: still positive at ₹1.46 lakh.

Scenarios (monthly saving)

Pessimistic	Plan	Observed
4.18 sigma	4.57 sigma	4.83 sigma
₹6.07 lakh	**₹8.40 lakh**	**₹8.95 lakh**

Bottom line: Errors start at the vendor and are caught late, so we fix them at submission; the saving stays positive under every flex we tested.

Improvement Results: Before vs After



1 Before: Baseline Performance (Aug – Sep 2026)

Audited invoices	1,669
Errors recorded	990
Defects per 100 audited invoices	59.32
DPMO (58 opportunities)	10,648
Sigma level	3.80

2 Key Improvement Interventions (targeted at highest-risk steps 5, 7 and 1–2)

PO matching validation
Automatic match with purchase order (PO).

→

Mandatory invoice fields
Checks for required fields and correct format.

→

Supporting document check
Validates vendor details and attachments.

→

Duplicate invoice check
Identifies and blocks duplicate submissions.

Focused on preventing errors at vendor submission and early validation.

3 After: Post-Improvement Performance (Oct 2026 – Feb 2027)

Audited invoices	1,505	-9.8% ↓
Errors recorded	38	-96.2% ↓
Defects per 100 audited invoices	2.52	-95.7% ↓
DPMO (58 opportunities)	452	-95.8% ↓
Sigma level	4.83	+1.03 ↑

4 Estimated Business Impact (using same assumptions)

Current state (baseline)

₹ 9.33 lakh estimated monthly error cost

₹ 1.12 crore estimated annual error cost

→

Post-improvement (projected)

₹ 0.36 lakh estimated monthly error cost

₹ 0.04 crore estimated annual error cost

~95.8% estimated reduction in monthly cost

5 Key Takeaways

1. Errors reduced from 990 to 38, a 96.2% decrease.
2. Defect rate improved from 59.32 to 2.52 per 100 invoices (95.7% reduction).
3. Sigma level increased from 3.80 to 4.83 (+1.03).
4. Estimated monthly error cost reduced from ₹9.33 lakh to ₹0.36 lakh (using the same assumptions).

Control Plan: Sustaining the Gains



1 Control Mechanisms

Control mechanism	What it checks	Frequency	Owner	Response / Action
Automated submission checks	• PO match • Mandatory fields • Attachments • Duplicate invoices	At submission (in real time)	IT / Process team	Reject or flag incomplete invoices with clear error message
Invoice quality audit	• Defects across 58 fields	Daily (as per sample plan)	Process owner (Invoice processing)	Record defects and investigate recurring patterns
Month-end monitoring	• Invoice volume • Defect rate • Peak-day load	Daily (with month-end review)	Process owner + Finance	Activate additional review or capacity measures
Performance dashboard	• DPMO • Defect rate • Top recurring categories • Rework effort	Weekly	Six Sigma consultant	Investigate adverse trends and assign actions
Management review	• Sustained performance against target • Open actions and risks	Monthly	CFO (sponsor) + Process owner	Review performance, assign corrective actions and close open items

2 Ongoing Monitoring Process

Capture → Invoices submitted through system (with validations)

Monitor → Track defect rate, DPMO and recurring categories

Flag deviation → If metrics exceed control limits

Investigate → Identify root cause and responsible step

Correct → Implement corrective action (e.g., system rule, process change)

Verify → Confirm improvement in next audit cycle

Continuous monitoring and improvement

3 Control Metrics and Targets

Defect rate
Maintain **≤ 2.52 per 100** invoices (or better)

Sigma level
Maintain **≥ 4.83** (or higher)

DPMO
Maintain **≤ 452** (or lower)

Monthly error cost
Maintain **≤ ₹0.36 lakh** (or lower) (using same assumptions)

4 Escalation Criteria

Within target
Continue monitoring

Marginal deviation
Investigate and take corrective action

Major deviation
Escalate to CFO and implement immediate fixes

Appendix: Formulas, Assumptions and Sources



1 Key Formulas (to rebuild in Excel)

Quality Metrics

Errors per audited invoice	Errors ÷ Samples checked
DPMO	Errors ÷ (Samples checked × 58) × 1,000,000
Sigma level	=NORMSINV(1 – DPMO ÷ 1,000,000) + 1.5
Weighted analyst touch time	20 + 25% × 12 = 23 minutes

Capacity and Cost Calculations

Daily capacity	18 × 8 × 60 ÷ 23 = 375 invoices (overtime days: 18 × 11 × 60 ÷ 23 = 516)
GST penalty exposure per month	Invoices × 35% × 5% × ₹5,000
Rework cost per month	Invoices × 35% × 12 ÷ 60 × ₹350

Statistical / Control Calculations

Control limit for a day with n audited invoices	0.025 + 3 × SQRT(0.025 ÷ n)
Two-proportion z-test	(p1 – p2) ÷ SQRT(p × (1 – p) × (1/n1 + 1/n2)), p = pooled rate, n = field checks

2 Assumptions Register

Assumption	Value	Basis	Used on	
GST penalty	₹5,000 per penalised invoice	Case statement ("up to")	1, 9, 11	Case data
Invoices with an error / penalised	35% / 5%	Cost Assumptions tab (case assumption)	9, 11	Team assumption
Analyst cost, overtime, exception time	₹350 an hour, ×1.5, 12 minutes	Cost Assumptions tab (case assumption)	8, 9	Team assumption
Share reaching exception handling	25% (capacity), 35% (cost)	Process Steps tab, Cost Assumptions tab	8, 9	Team assumption
Invoices a month	8,332	Measure average, inside the 7,000 to 9,100 range	9, 11	Case data
Peak day	400 or more invoices	Team definition (about 1.25 times average)	1, 5	Team assumption
25 Aug	Samples excluded, volume kept	Team treatment (Slide 3)	3 to 11	Team assumption
Sigma shift	1.5	Standard convention	4, 11	Team assumption
Error reason mapped to process step	See Slide 8	Team judgement	8	Team assumption
Savings follow the error reduction	Proportional, worst case also shown	Team assumption	1, 10, 11	Team assumption
Six-month programme placement	Oct 2026 to Mar 2027	Fits the 1 Mar 2027 audit start	12	Team assumption
Goal and control thresholds	90% DPMO cut; limits on Slide 12	Team-set	2, 12	Team-defined threshold
Three-way match tolerance	Set by Finance	Not given in the case	10	Team assumption

3 Sources

Organisers' Case PDF and workbook

Please refer this google drive link

CBIC

Sage

Note: Case-provided values are distinguished from team assumptions. Calculated results depend on the stated formulas and assumptions. Risk ratings are the team's judgement of likelihood and effect, not calculated.